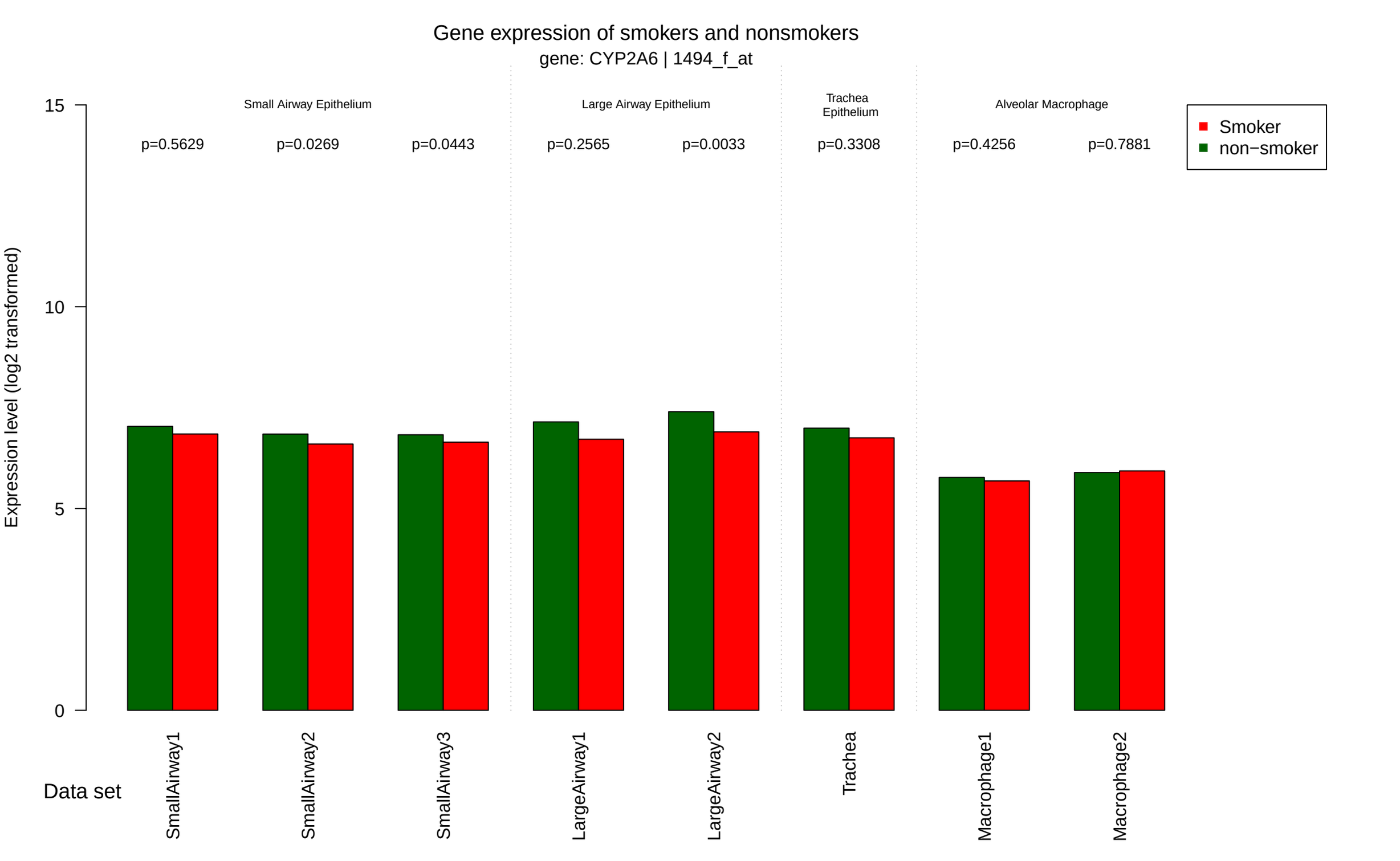
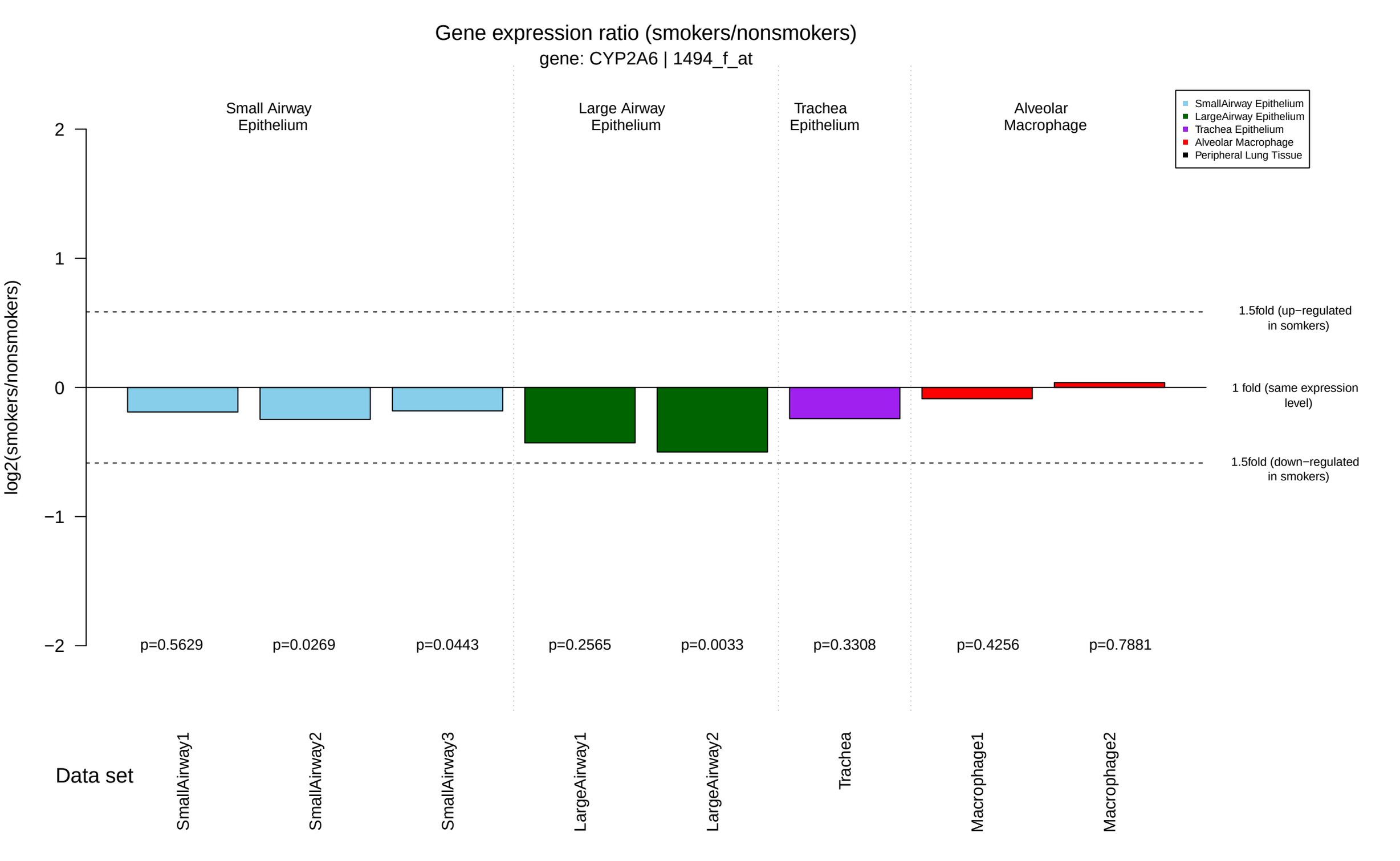
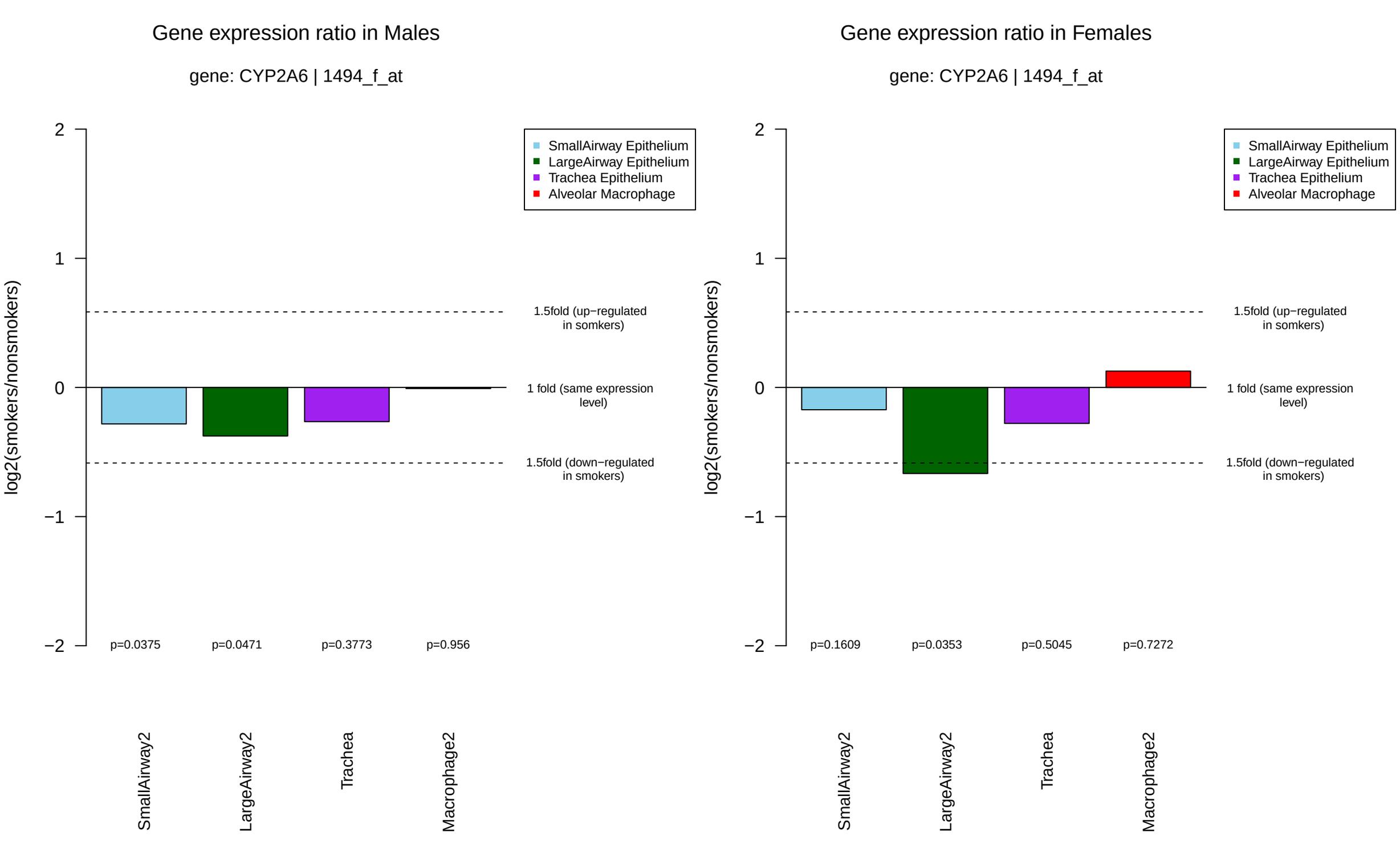
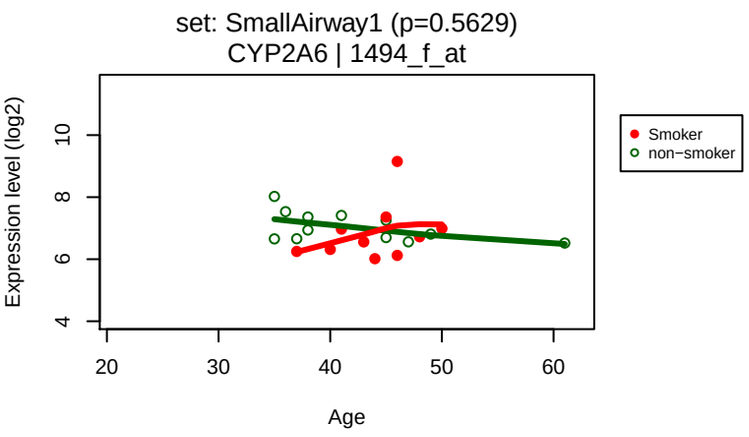
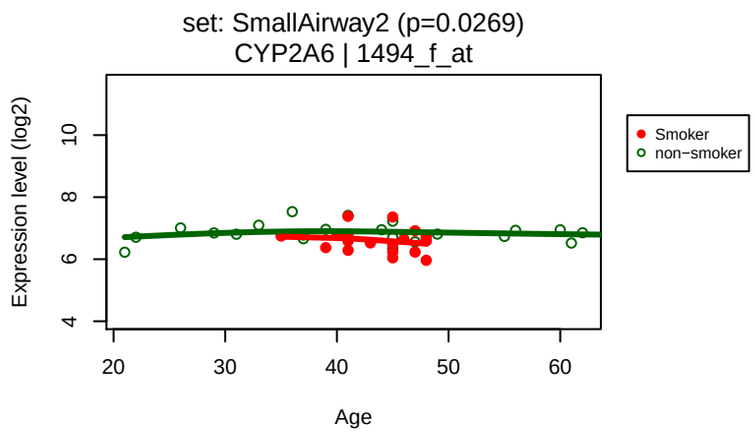
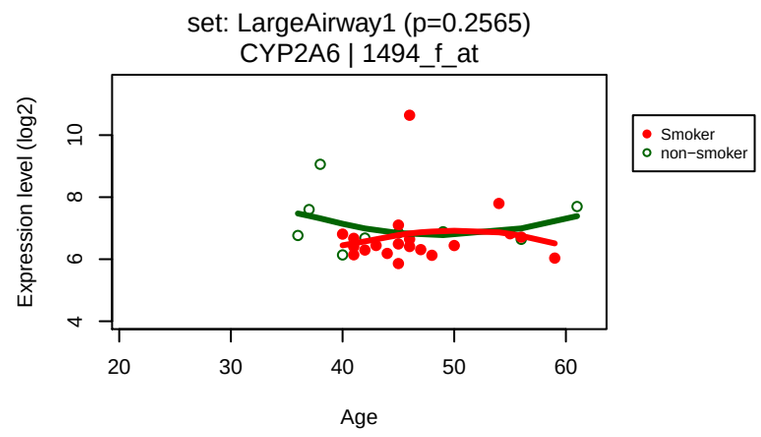

Gene expression of smokers and nonsmokers
gene: CYP2A6 | 1494_f_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5629
p=0.0269
p=0.0443
p=0.2565
p=0.0033
p=0.3308
p=0.4256
p=0.7881
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

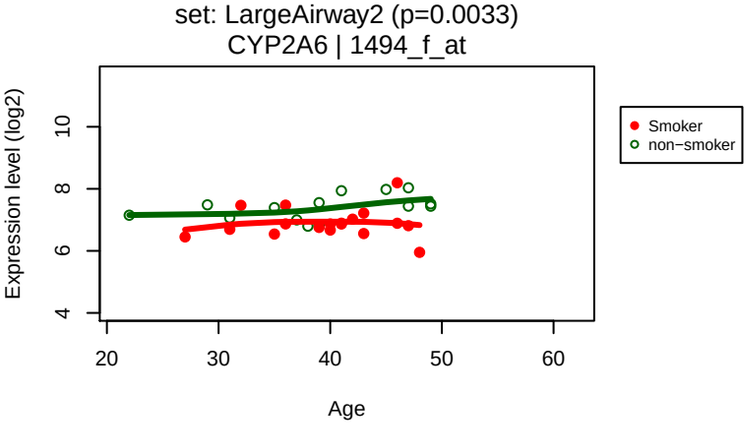

Gene expression ratio (smokers/nonsmokers)
gene: CYP2A6 | 1494_f_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5629
p=0.0269
p=0.0443
p=0.2565
p=0.0033
p=0.3308
p=0.4256
p=0.7881
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

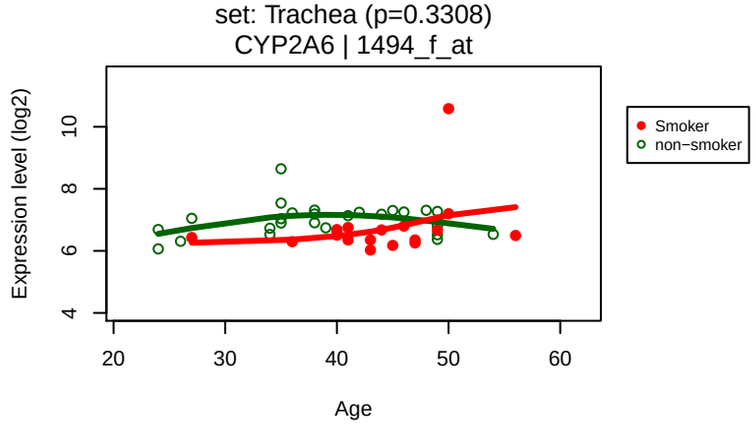

Gene expression ratio in Males
gene: CYP2A6 | 1494_f_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0375
p=0.0471
p=0.3773
p=0.956
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2A6 | 1494_f_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1609
p=0.0353
p=0.5045
p=0.7272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

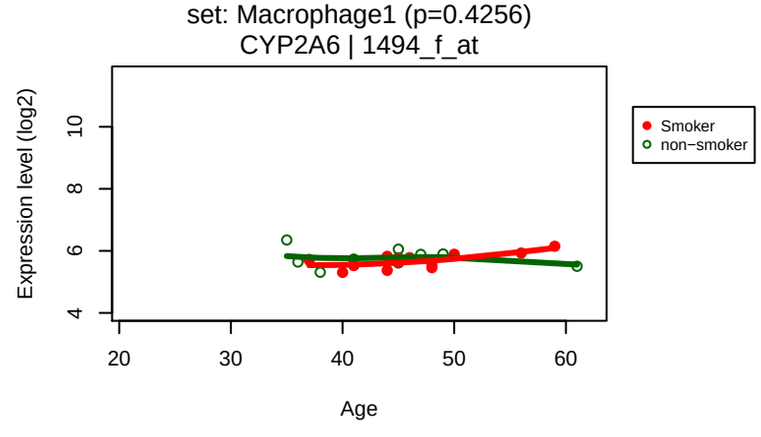

set: SmallAirway1 (p=0.5629)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker

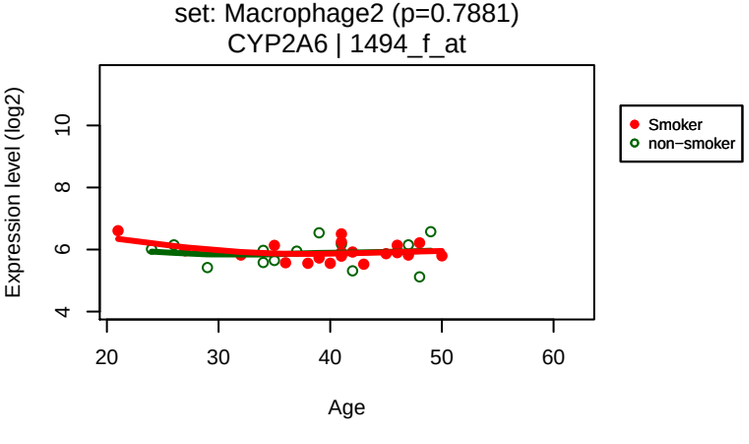

set: SmallAirway2 (p=0.0269)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker

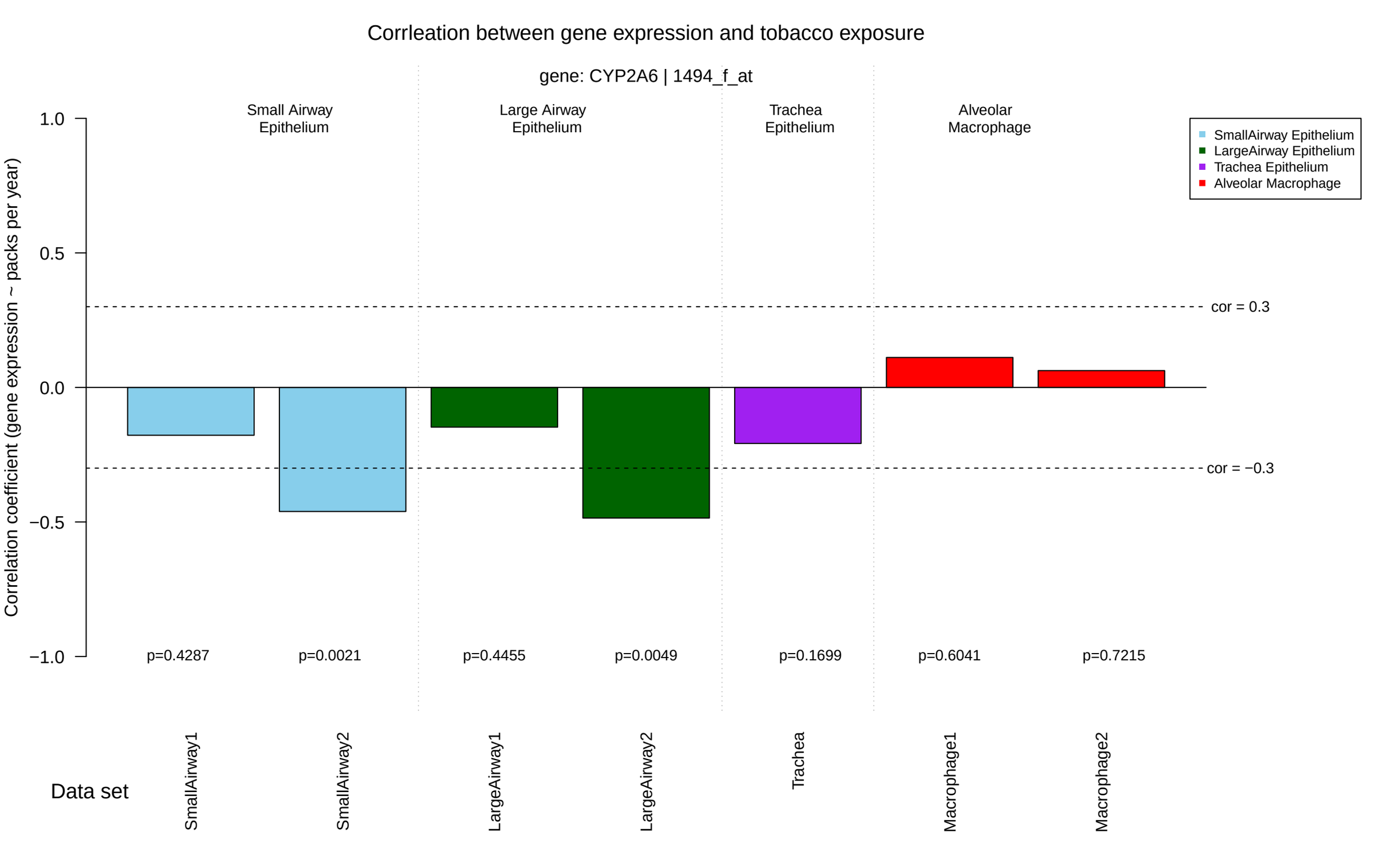

set: LargeAirway1 (p=0.2565)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0033)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3308)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4256)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7881)
CYP2A6 | 1494_f_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2A6 | 1494_f_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4287
p=0.0021
p=0.4455
p=0.0049
p=0.1699
p=0.6041
p=0.7215
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2